

ZEROTRUST FORENSICS

Digital Forensics Case Management & Cryptographic Custody System

Comprehensive Mentor Presentation Guide, Module Breakdown & Verification Manual

1. Executive Overview & Problem Statement

Digital evidence (memory dumps, disk images, log files) collected during cybercrime investigations faces critical vulnerabilities:

- Undetected File Alteration: Evidence files can be modified on disk. Without continuous fingerprinting, changes go unnoticed.
- Weak Custody Accountability: Traditional systems record custody as simple text edits, failing to prove who authorized a transfer.
- Editable Audit History: Standard database logs can be altered or deleted by privileged database administrators.
- Superficial Role Controls: Hiding UI buttons fails if backend API endpoints do not strictly enforce server-side authorization.

Solution: ZeroTrust Forensics enforces continuous verification across all workflows via raw SHA-256 fingerprinting, dual Ed25519 digital signatures, append-only hash-linked audit ledgers, and AES-256-GCM encryption.

2. How to Present to your Mentor (Elevator Pitch & Novelty)

Key Talking Points for Evaluators:

- **Bi-Directional Tamper Detection (File Level + Chain Level)**

Blockchains only verify that a hash string was logged, but cannot detect if a 20GB physical file on disk was modified. Our system computes fresh SHA-256 hashes off raw file bytes on demand, raising FILE_TAMPERING_DETECTED alerts instantly.

- **Dual-Party Cryptographic Ed25519 Custody Approval**

Unlike unilateral blockchain pushes, custody transfers stay PENDING until BOTH Sender and Intended Receiver supply independent Ed25519 signatures verified via server-side public key cryptography.

- **Lightweight Hash-Linked Ledger (Zero Gas Fees)**

Provides the exact tamper-evident mathematical security of a blockchain hash chain without P2P node latency or expensive gas fees.

- **Interactive Live Anomaly & Tamper Simulator**

Includes built-in Simulate Tampering buttons allowing mentors to flip 1 byte on disk live and watch the system catch the violation.

3. Module-by-Module Technical Breakdown

Module 1: Authentication & Server-Enforced RBAC

Passwords hashed using scrypt + 16-byte random salts + timingSafeEqual comparison. Enforces 4 roles: Admin, Investigator, Lab Analyst, and Auditor. Server endpoints validate session cookies on every request.

Module 2: Forensic Case Management Directory

Manages investigation dockets with classification levels (SECRET, CONFIDENTIAL, RESTRICTED, UNCLASSIFIED). Links evidence artifacts, lead investigator details, and audit history.

Module 3: Evidence Vault & SHA-256 Fingerprinting

Computes genuine SHA-256 hash on intake from raw file bytes. Re-verifies disk files on demand. Includes AES-256-GCM metadata encryption and printable Cryptographic Evidence Receipts.

Module 4: Chain of Custody & Ed25519 Dual Signatures

Auto-generates Ed25519 keypairs for all users. Custody transfers require independent digital signatures from both Sender and Receiver before updating current custodian.

Module 5: Append-Only Hash-Linked Audit Ledger

Logs every action in a linked chain: Current_Hash = SHA-256(Seq | Timestamp | Actor | Action | Resource | Details | Prev_Hash). Includes chain verification and repair tools.

Module 6: Security Alerts Center

Monitors anomalies (file tamper alerts, invalid signature attempts, audit chain breaks) with severity levels (CRITICAL, HIGH, MEDIUM, LOW) and auditor resolution notes.

4. Step-by-Step Live Working Verification Checklist

Follow this exact 10-step walkthrough during your mentor presentation:

1. Open <http://localhost:3001> in your browser.
2. Click "Investigator" on the Quick Login Bar to sign in as Det. Marcus Vance.
3. Go to "Cases Directory" -> Click "Create New Case Docket".
4. Go to "Evidence Vault" -> Click "Upload & Fingerprint Evidence". Upload any file.
5. Observe the genuine 64-character SHA-256 hash generated automatically.
6. Click "Verify Hash" -> Confirm green INTACT status.
7. Click "Simulate Tamper" -> Click "Verify Hash" -> Watch status turn RED TAMPERED and raise a CRITICAL Alert!
8. Click "Restore File" -> Re-verify to confirm return to INTACT.
9. Go to "Custody Transfers" -> Initiate transfer to Elena Rostova (Lab Analyst). Observe Sender Ed25519 signature.
10. Switch to "Lab Analyst" account -> Click "Sign & Accept Custody" -> Observe dual signature validation & custodian update.

5. Demo Credentials Reference

- Admin: admin@zerotrust.local / AdminPass123! (Dr. Sarah Chen)
- Investigator: investigator@zerotrust.local / InvestigatorPass123! (Det. Marcus Vance)
- Lab Analyst: analyst@zerotrust.local / AnalystPass123! (Elena Rostova)
- Auditor: auditor@zerotrust.local / AuditorPass123! (James Sterling)